CSEC Physics

January 2020 – Paper 2

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SECTION A

Answer ALL questions.

1. A sprinter of mass 55 kg was running a 100 m race. Her velocity was measured at different intervals of a 11.0 s period. The results are recorded in Table 1.

TABLE 1: RESULTS

Velocity, v/ms^{-1}	0.00	1.20	3.75	8.75	10.00	10.00
Time, t/s	0.0	1.0	3.0	7.0	8.0	11.0

(a) (i) Define the term 'velocity'. (2 marks)

Velocity is defined as the rate of change of displacement.

$$v = \frac{\Delta s}{\Delta t}$$

where v = velocity (ms^{-1})

Δs = change in displacement (m)

Δt = change in time (s)

(ii) Complete the following statement. (1 mark)

Acceleration is defined as the rate of change of velocity

(b) Using the grid provided **on page 5**, plot a graph of velocity, v/ms^{-1} against time, t/s. (8 marks)

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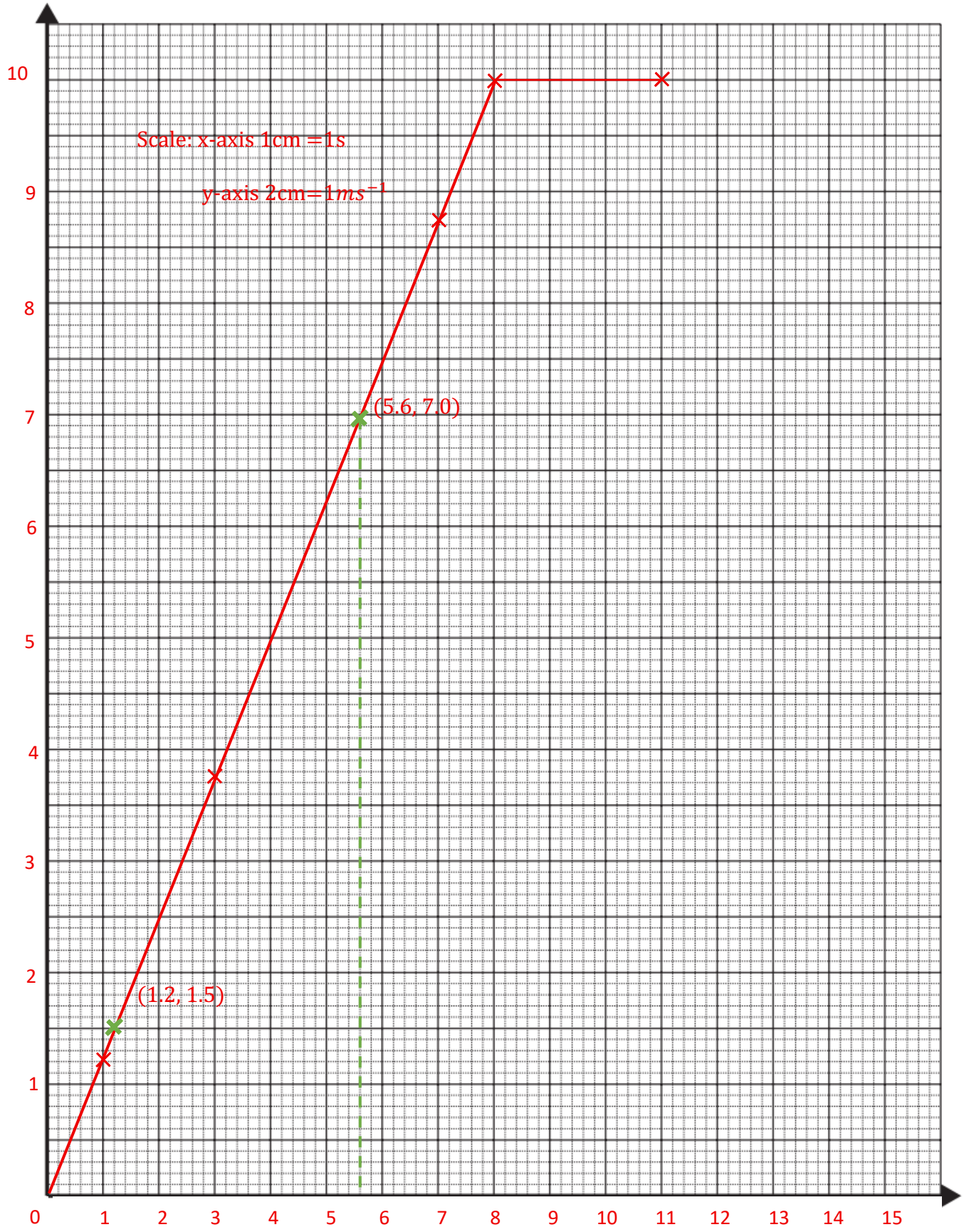

Figure 1. Graph of velocity, v/ms^{-1}, against time, t/s

(c) Determine the slope of the graph over the first 7.0 s of the race. [4]

$$Gradient = \frac{y_2 - y_1}{x_2 - x_1}$$

Using points (5.6, 7.0) and (1.2, 1.5) from Graph

$$Gradient = \frac{7.0-1.5}{5.6-1.2}$$

$$= 1.25\ ms^{-2}$$

(d) (i) State the physical quantity relating to the motion of the sprinter which the slope of the graph represents. [1]

Acceleration

(ii) Describe the sprinter's motion from 8 s to 10 s. [1]

For the period 8s to 10 seconds, the sprinter is running at a consistent velocity since the gradient at that point is ZERO, that means that acceleration is zero, hence the sprinter continues at a fixed velocity.

(e) Calculate the resultant force acting on the sprinter after 7.0 s. [3]

$$F = ma$$

$$= (55)(1.25)$$

$$= 68.75N$$

(f) Using the graph drawn **on page 5**, determine how far from the finish line the sprinter would be after 11.0 s. [5]

Distance = Area under graph

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$$From\ 0s - 8s:\ Area = \frac{1}{2}(8)(10)$$

$$= 40m$$

$$From\ 8s - 11s:\ Area = (10)(3)$$

$$= 30m$$

$$Total\ Distance\ from\ 0s - 11s = 40 + 30$$

$$= 70m$$

$$Since\ it's\ a\ 100\ metre\ race\ distance\ remaining:$$

$$= 100m - 70m$$

$$= 30m$$

Total: 25 marks

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2. (a) Define EACH of the following terms:

(i) Specific heat capacity [3]

The specific heat capacity of a substance is the amount of heat energy required to raise the temperature of 1 kg of the substance by 1 °C (or 1 K) without a change of state.

(ii) Specific latent heat of fusion of ice [3]

The specific latent heat of fusion is the amount of heat energy required to change 1 kg of ice at 0 °C into water at 0 °C without any change in temperature.

(b) A block of ice, which has a mass of 2.5 kg and a temperature of $-10\ °C$, is placed in a container.

Specific heat capacity of water $= 4.2 \times 10^3\ Jkg^{-1}(°C)^{-1}$

Specific heat capacity of ice $= 2.1 \times 10^3\ Jkg^{-1}(°C)^{-1}$

Specific latent heat of fusion of ice $= 3.3 \times 10^5\ Jkg^{-1}$

Calculate

(i) the heat energy required to change the temperature of the ice at $-10\ °C$ to ice at $0\ °C$ [3]

$$E = mc\Delta\theta$$

$$= (2.5)(2.1 \times 10^3)\big(0 - (-10)\big)$$

$$= 52500J$$

(ii) the heat energy required to convert all the ice at $0\ °C$ to water at $0\ °C$. [2]

$$E = ml_f$$

$$= (2.5)(3.3 \times 10^5)$$

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$$= 8.25 \times 10^5 J$$

(c) The water in the container is now heated to a temperature of 80 °C by an electrical heater, in 600 seconds.

(i) Calculate the power output of the heater. [3]

$E = mc\Delta\theta$

$= (2.5)(4.2 \times 10^3)(80 - 0)$

$= 8.4 \times 10^5 J$

(assuming no energy is lost)

Energy required to heat the water = Energy required by heater

$$E = Pt$$

$$P = \frac{E}{t}$$

$$P = \frac{8.4 \times 10^5}{600}$$

$$= 1400W$$

(ii) State ONE assumption which must be taken into account in (c)(i). [1]

All the energy supplied by the heater is absorbed by the water with no heat/energy loss.

Total: 15 marks

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3. (a) (i) Define the term 'linear momentum'. [1]

Linear momentum is defined as the product of the mass of an object and its velocity.

Equation: $p = mv$

where p = momentum $(kgms^{-1})$

m = mass (kg)

v = velocity (ms^{-1})

(ii) State the SI unit for linear momentum. [1]

The SI unit for linear momentum is $kgms^{-1}$.

(b) (i) State the law of conservation of linear momentum. [3]

The law of conservation of linear momentum states that, in the absence of external forces, the total momentum of a system of bodies is constant.

(ii) State TWO real-life examples that demonstrate the law of conservation of linear momentum. [2]

Two real-life examples are:

- a game of marbles where two marbles hit each other
- pool table (billiards) where a player hits a ball with another ball.

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(c) Trolley A and Trolley B collided head-on. Trolley A of mass 72.0 kg was moving towards the west at 5.50 ms^{-1}. Trolley B of mass 69.4 kg was moving towards the east at 4.50 ms^{-1}.

Calculate

(i) the initial momentum of Trolley A [2]

Initial momentum of Trolley A:

$$p = mv$$

$$= 72 \times 5.5$$

$$= 396\ kgms^{-1} \quad \text{(to the west direction)}$$

(ii) the initial momentum of Trolley B [2]

Initial momentum of Trolley B:

$$p = mv$$

$$= 69.4 \times 4.5$$

$$= 312.3\ kgms^{-1} \quad \text{(to the east direction)}$$

(d) Assuming the trolleys locked during the collision and that they moved off together in one direction, determine the speed and the direction of the joined trolleys immediately after the collision. [4]

Let East be positive.

By the law of conservation of linear momentum,

Total momentum before collision = Total momentum after collision

$$p_1 + p_2 = p_3$$

$$m_1v_1 + m_2v_2 = m_3v_3$$

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$$312.3 - 396 = (141.4)v_3$$

$$-83.7 = (141.4)v_3$$

$$v_3 = \frac{-83.7}{141.4}$$

$$v_3 = -0.592\ ms^{-1}$$

∴ The speed and direction of the joined trolleys after the collision is $0.592\ ms^{-1}$ to the west.

Total: 15 marks

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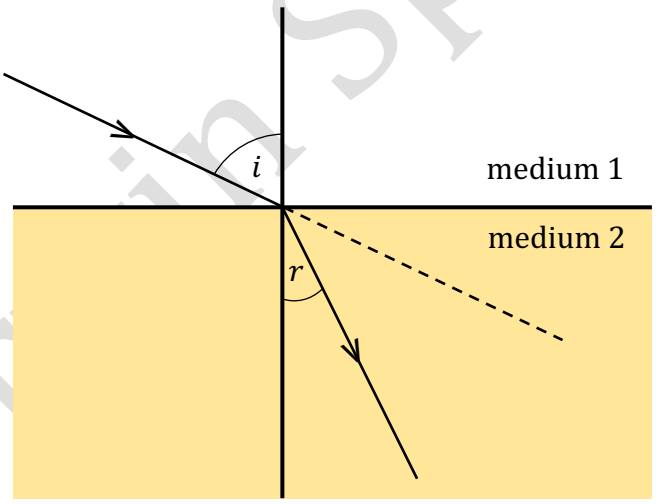
SECTION B

Answer ALL questions.

4. (a) (i) With the aid of a labelled diagram, state the TWO laws of refraction. [4]

Laws of Refraction:

1. The incident ray, the refracted ray and the normal, at the point of incidence, all lie on the same plane.
2. The ratio $\frac{\sin i}{\sin r}$ is a constant for a given pair of media, where i is the angle of incidence and r is the angle of refraction. This is Snell's law.



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(ii) When a diver under the sea shines a beam of light towards the surface of the water, the light does not pass through the water-air boundary into the air above but reflects into the sea.

State the name of the phenomenon which occurs **and** briefly explain why it occurs. [2]

The name of the phenomenon is total internal reflection and it occurs when light travels from a more optically dense medium to a less optically dense medium, the angle of incidence of the light ray is greater than the critical angle the light ray is completely reflected back into the sea.

(b) Another diver shines a beam of light into the water to see some fishes as shown in Figure 2.

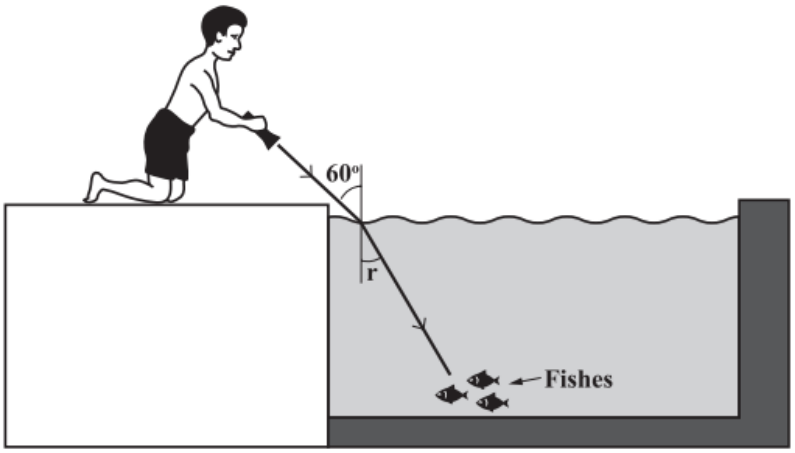

Figure 2. Diagram showing refraction

Given that the index of refraction for the water is 1.33 and the angle of incidence, $i = 60°$, calculate the angle of refraction, r. [3]

$$\frac{\sin i}{\sin r} = n$$

$$\frac{\sin 60}{\sin r} = \frac{1.33}{1}$$

$$\sin r = \frac{\sin 60}{1.33}$$

$$r = \sin^{-1}\left(\frac{\sin 60}{1.33}\right)$$

$$r = 40.6° \qquad \text{(to 1 decimal place)}$$

∴ The angle of refraction is 40.6°

(c) An object 2 cm in height, is placed 10 cm in front of a converging lens with a focal length, f, of 6 cm. Calculate

(i) the image distance, v. [4]

$f = 6 \qquad u = 10$

$$\frac{1}{6} = \frac{1}{v} + \frac{1}{10}$$

$$\frac{1}{v} = \frac{1}{6} - \frac{1}{10}$$

$$\frac{1}{v} = \frac{5-3}{30} = \frac{2}{30} = \frac{1}{15}$$

$$v = 15 \text{ cm}$$

(ii) the linear magnification, m. [2]

$$Mag = \frac{v}{u}$$

$$Mag = \frac{15}{10}$$

$$= 1.5$$

Total: 15 marks

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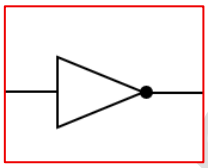
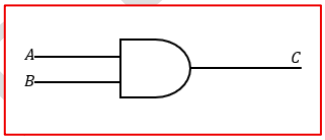
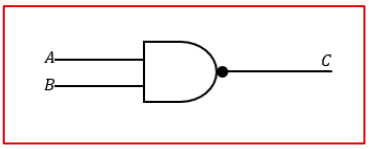
5. (a) Complete Table 2 by inserting the missing logic gates, symbols and functions. [6]

TABLE 2: LOGIC GATES, SYMBOLS AND FUNCTIONS

Logic Gate	Symbol	Function
NOT		Changes logic 0 to logic 1 **(and Changes logic 1 to logic 0)**
AND		Produces a logic 1 when both A and B are 1
OR		Produces a logic 1 when either A or B or both are 1
NAND		Produce a logic 1 only when A and B are logic 0 **(NOTE: Error in CSEC exam paper this describes a NOR gate.)**

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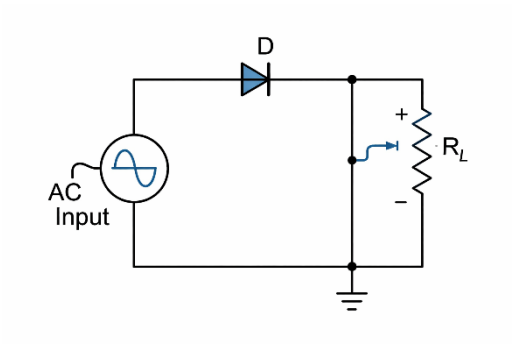
(b) (i) With the aid of a circuit diagram, describe how an alternating current (a.c.) can be half-wave rectified to produce a direct current across a resistor (load). [5]

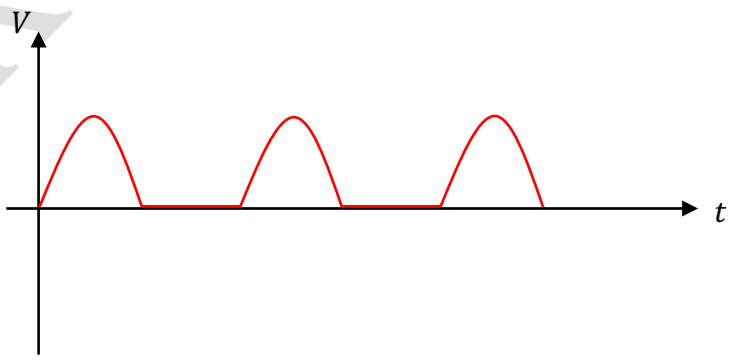


When a diode is placed in the circuit diagram, only half of the wave would be allowed through.

This is called half-wave rectification.

Half-wave rectification is the process of converting ac to dc by preventing one half of each cycle from being applied to the load.

(ii) Sketch the voltage-time graph to show the expected output across the load in (b)(i). [1]



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(c) Draw a voltage-time graph showing an a.c. source with a frequency of 5 Hz and a peak voltage of 12 V. [3]

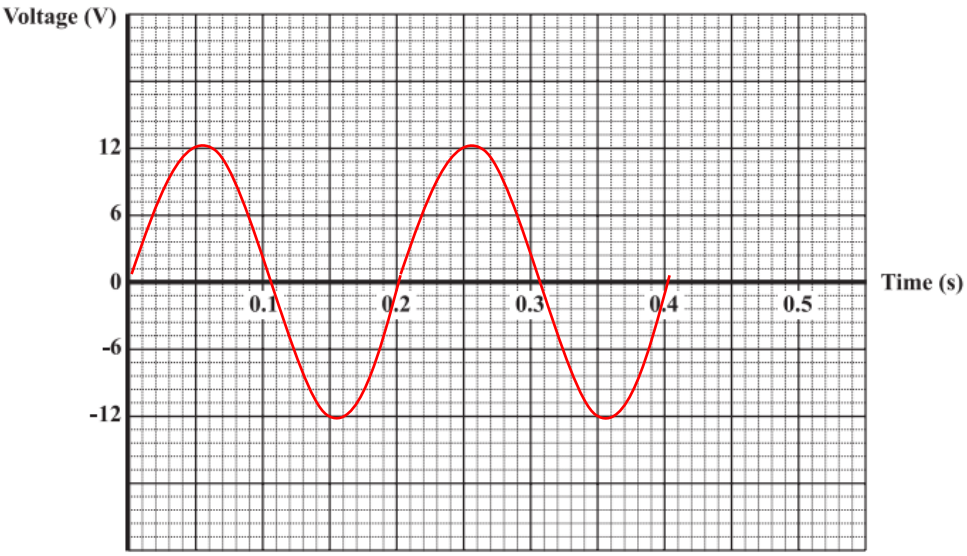

Figure 4. Voltage–time graph showing an a.c. source

$$f = \frac{1}{Period}$$

$$Period = \frac{1}{5}$$

$$= 0.2s$$

Total: 15 marks

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6. (a) Three emissions, α, β, γ, from a radioactive source are subjected to a uniform magnetic field perpendicular to their path, as shown in Figure 5.

(i) Draw the three rays to represent the path of EACH of the three emissions, α, β, γ, on Figure 5, as they pass through the magnetic field. [3]

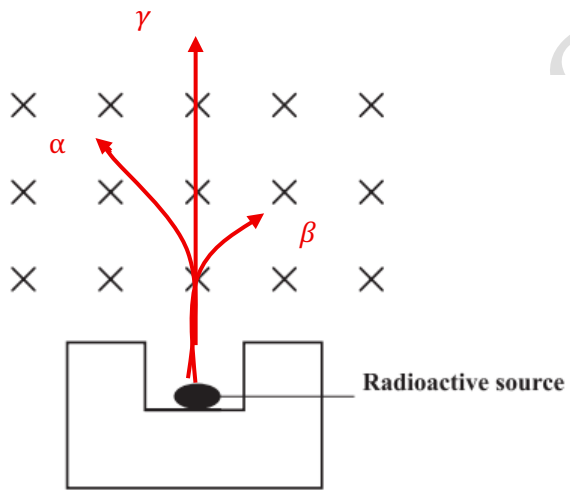

Figure 5. Radioactive source in a uniform magnetic field

(ii) The magnetic field is replaced by an electric field as shown in Figure 6. Draw the three rays to represent the path of EACH of the three emissions, α, β, γ, on Figure 6, as they pass through the electric field. [3]

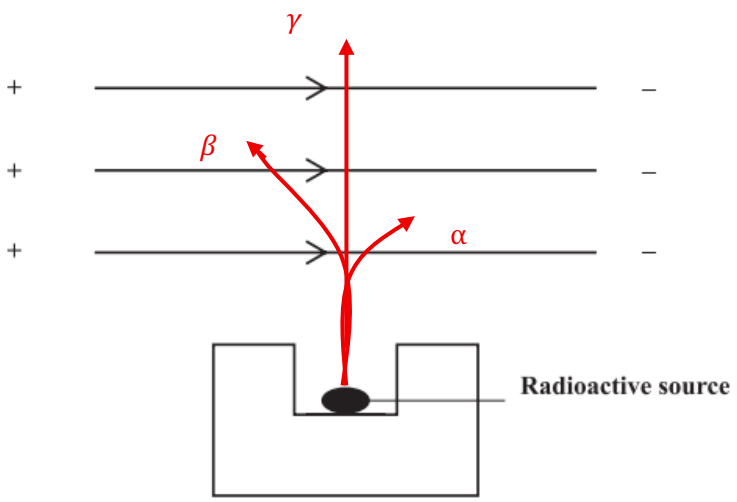

Figure 6. Radioactive source in an electric field

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(b) An unstable isotope of lead, ${}^{210}_{82}Pb$, undergoes exactly two sequential beta particle decays followed by an alpha particle decay to becomes stable lead, ${}^{206}_{82}Pb$.

With reference to the information in Table 3, write the nuclear equations that will result in the formation of stable lead, ${}^{206}_{82}Pb$ from ${}^{210}_{82}Pb$. [6]

TABLE 3: ELEMENTS AND ATOMIC NUMBERS

Element	Atomic Number
Bi	83
Po	84

First beta decay equation: ${}^{210}_{82}Pb \rightarrow {}^{210}_{83}Bi + {}^{0}_{-1}e^{-1}$

Second beta decay equation: ${}^{210}_{83}Bi \rightarrow {}^{210}_{84}Po + {}^{0}_{-1}e^{-1}$

Alpha decay equation: ${}^{210}_{84}Po \rightarrow {}^{206}_{82}Pb + {}^{4}_{2}\alpha$

(c) Calculate the energy given off in a nuclear reaction if the change in mass is $0.3018\ u$.

$(u = 1.66 \times 10^{-27}\ kg, c = 3 \times 10^{8}\ ms^{-1})$ [3]

Mass defect = 0.3018 u

1 u = 1.66×10^{-27} kg

c = 3.0×10^{8} m s^{-1}

Mass defect in kg = $0.3018 \times 1.66 \times 10^{-27} = 5.00988 \times 10^{-28}$ kg

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$$E = mc^2$$

$$E = 5.00988 \times 10^{-28} \times (3 \times 10^8)^2$$

$$E = 5.00988 \times 10^{-28} \times 9 \times 10^{16}$$

$$E = 4.51 \times 10^{-11}\ \text{J}$$

Total: 15 marks

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END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST

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